



→ crash-prone) under the name of Silvermont. These fresh crop of low-powered chips (the company claims power consumption is only a fifth of the competition) coupled with Intel's 22nm manufacturing process promises to be a compelling combination. By producing these as SoCs (System on Chips) rather than discrete CPUs, Intel has made its intentions very clear.

ARMs business model is unique in that it generates almost all of its revenue by licensing its chip designs to companies like Samsung, Qualcomm, Fairchild and Texas Instruments. Intel, however, has the distinct competitive advantage of owning possibly the most sophisticated chip fabrication process in the world, so the next few years might see a fresh resurgence of mobile devices using Intel's processors. Not to be left behind, ARM is rumoured to be moving to a 16nm process architecture in the next 2-3 years. (A lower transistor width leads to lower power consumption, and a processor typically consists of millions of such transistors). This race to the bottom in transistor widths can only be a good thing for end users - since it translates into faster, more power efficient chips that can be mass-produced at a cheaper price.

What Intel has going for it is its vast experience and the end-to-end chip design-manufacture-sell technology. If the Silvermont SoCs live up to their promise of extreme power-efficiency, Intel will gain a technological advantage as well. ARM (along with its partners like Samsung, Qualcomm, NVidia and TI) have chips that currently rule the roost. Most mobile OSes have been written atop the ARM architecture, as Android's x86 port is yet to gain significant traction while iOS is written to run exclusively on Apple's custom ARM chips. Whether Intel can manage to break into this segment, and whether other competitors can continue to hold their own - makes for a very interesting battle.

3D Printing

3D printing, is exactly what it sounds like. It is a manufacturing process of creating a 3-dimensional object using a 2-dimensional reference design, using an additive



ARM's dominance is chiefly due to their low-power chip designs.

process where a material like metal, polymer, ceramic or plastic is deposited in the form of thin, successive layers. This is in contrast to the traditional manufacturing processes like drilling or filing, which are subtractive processes that remove excess material from a block of material to manufacture the final product. The result is a product that can be quickly produced, at a reasonably low cost with a high degree of accuracy. These qualities make it an ideal tool for rapid prototyping in almost every industry. Costs of 3D printers have come down significantly in the last few years, and department stores like Staples have started stocking them as a result, making them accessible to the masses.

While some of the more interesting use cases for a 3D printer have included prosthetic devices such as a prosthetic limb, ear or even a set of dentures, a working gun, musical instruments or even 3D photographs, the one application that has impressed us most is 3D food. Imagine being able to design your own pizza on a computer screen and have it 'manufactured' to your specifications! These 'food printers' would use cartridges that contained powdered forms of carbohydrates,



Intel's Silvermont series holds a lot of promise.

proteins and vitamins that would supply the requisite nutrients in the food. This application is being actively explored by a company called Systems & Materials Research Corporation, a company that has received funding from NASA who seem to be very interested in the implications of this. If successful, these 3D printers could provide an innovative solution to the problem of transporting, storing and preserving food in outer space. The food cartridges that the proposed printer will use, are durable and the ingredients can stay fresh for almost 30 years, as all the moisture is drained from the cartridges prior to being loaded. Anjan Contractor, the engineer pioneering the work in this field, says that pizza is one of the simplest foods to "print" this way, as it has an inherently layered structure, and the printer would only have to extrude one layer of material at a time.

Closer to earth, one reason scientists are excited by this application is that it provides an elegant and practical solution to the problem of food shortage and malnutrition in developing countries. While traditional methods of food cultivation such as farming, fishing or rearing poultry are bound to fall short of the demand, 3D printers can be deployed in bulk to produce food that contains carefully measured levels of nutrients in a hygienic environment.

By using innovative (if slightly audacious) sources of protein such as insects, grass and algae to supplement traditional sources, scientists can ensure a steady supply of raw materials. Furthermore, the food's nutritional content can be tailored to suit the dietary requirements of different individuals.

3D printing may not be a new technology, but recent advances in computing power have given it enormous potential. Prosthetic devices such as bionic arms, ears, legs have aided in saving the lives of many people in recent years. Doctors also successfully replaced a man's entire face using a 3D printed prosthetic after the face was badly damaged by a tumour. On the personal electronics front, it could